Calculus I
Exam 5
2 December 2026

- show all necessary work
- for full credit, answers should be simplified algebraically
- answer questions in order on blank sheets of paper with each question beginning below the previous one or on a new page; you may feel it is best to write one question per page; staple all pages when you are done
- the minimum score you have already earned is next to each problem
- there are some useful formulas on the back of this page

- $\frac{d}{dx}[c] = 0$
- $\frac{d}{dx}[x^n] = nx^{n-1}$
- $\frac{d}{dx}[a^x] = a^x ln(a)$
- $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$
- $\frac{d}{dx}[\sin(x)] = \cos(x)$
- $\frac{d}{dx}[\cos(x)] = -\sin(x)$
- $\frac{d}{dx}[\tan(x)] = \sec^2(x)$
- $\frac{d}{dx}[\cot(x)] = -\csc^2(x)$
- $\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$
- $\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$
- $\frac{d}{dx}[\arcsin(x)] = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}[\arccos(x)] = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}[\arctan(x)] = \frac{1}{x^2+1}$
- $\frac{d}{dx}[kf(x)] = k\frac{d}{dx}[f(x)]$
- $\frac{d}{dx}[f(x)+g(x)] = \frac{df}{dx} + \frac{dg}{dx}$
- $\frac{d}{dx}[f(x)g(x)] = f\frac{dg}{dx} + g\frac{df}{dx}$
- $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g\frac{df}{dx} - f\frac{dg}{dx}}{g^2}$
- $\frac{d}{dx}[f(g(x))] = f'(g(x))\,g'(x)$
- $\frac{d}{dx} = \frac{dy}{dx}\frac{d}{dy}$
- $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}$
- $L_n = \Delta x \sum_{i=0}^{n-1} f(x_i)$
- $R_n = \Delta x \sum_{i=1}^{n} f(x_i)$
- $M_n = \Delta x \sum_{i=0}^{n-1} f\left(\frac{x_i + x_{i+1}}{2}\right)$
- $\int_a^a f(x)dx = 0$
- $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$
- $\int_a^b f(x)dx = -\int_b^a f(x)dx$
- $\int_a^b kf(x)dx = k\int_a^b f(x)dx$
- $\int_a^b f(x) + g(x)dx = \int_a^b f(x)dx + \int_a^b g(x)dx$
- $\frac{\int_a^b f(x)dx}{b-a}$
- $\int_a^b f(x)dx = F(b) - F(a)$

Name:

1. Evaluate each of the following.

   (a) **/5** $\dfrac{d}{dx}\left[\displaystyle\int_2^x \ln\left(1+t^2\right)dt\right]$

   (b) **/5** $\displaystyle\int_{-1}^x \frac{d}{dt}\left[\frac{t^5}{1+t^6}\right]dt$

   (c) **/5** $\dfrac{d}{dx}\left[\displaystyle\int_x^3 e^{t^2}dt\right]$

   (d) **/5** $\dfrac{d}{dx}\left[\displaystyle\int_{-2}^{4x} \cos\left(t^3\right)dt\right]$

2. Evaluate each integral. Hint: remember you can check your answer by differentiating.

   (a) **/5** $\displaystyle\int \cos\left(8-3x\right)dx$

   (b) **/5** $\displaystyle\int \frac{dx}{16x^2+1}$

   (c) **/5** $\displaystyle\int \frac{x^3dx}{2x^4+9}$

   (d) **/5** $\displaystyle\int_{\frac{2}{\pi}}^{\frac{4}{\pi}} \frac{\sin\left(\frac{1}{x}\right)}{x^2}dx$